Process Variation Mitigation Techniques

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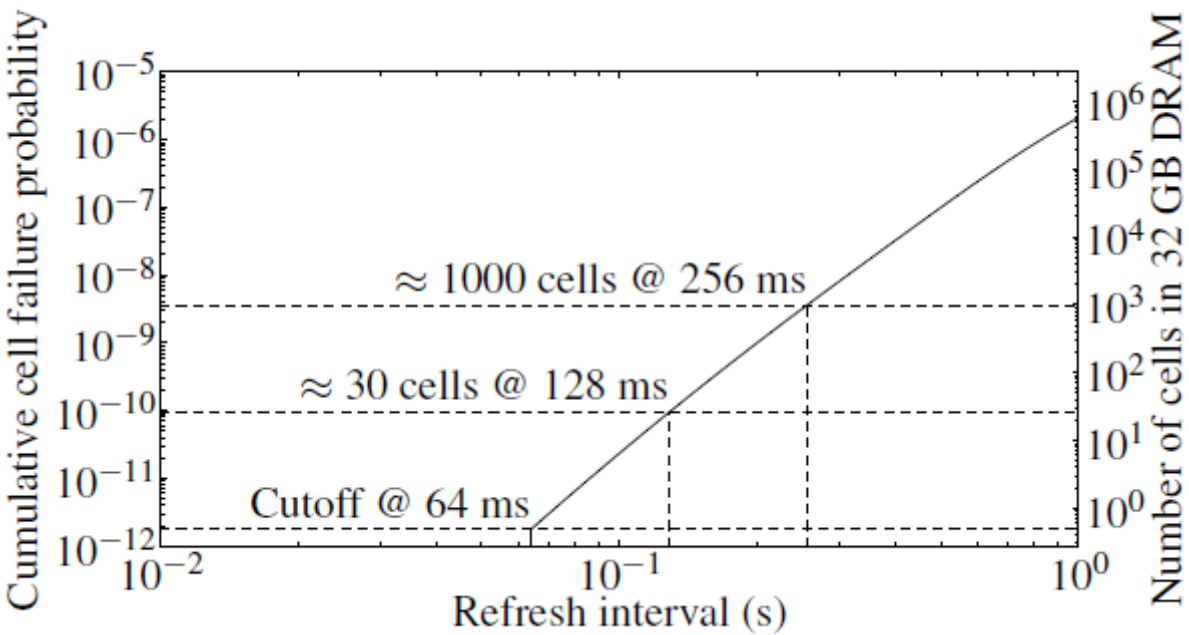
Process Variation: A Crucial Reliability Challenge

- **Process Variation (PV):** leads to significant variability in reliability, performance and power profile

- PV makes it difficult to estimate device parameters accurately and design reliability solutions

Example: Variation in DRAM cell retention times

- PV can cause large variation in reliability profile



Liu et al., "RAIDR: Retention-aware intelligent DRAM refresh", ISCA, 2012.

Example: Frequency and leakage variation across tiles of a chip

M_0 f = 1.186 I_{leak} = 4.771	P_1 f = 1.191 I_{leak} = 4.974	P_0 f = 1.195 I_{leak} = 5.161	M_1 f = 1.199 I_{leak} = 5.329
P_{14} f = 1.154 I_{leak} = 3.628	M_{15} f = 1.158 I_{leak} = 3.753	M_{14} f = 1.162 I_{leak} = 3.866	P_{15} f = 1.165 I_{leak} = 3.964
P_2 f = 1.125 I_{leak} = 2.812	M_3 f = 1.128 I_{leak} = 2.890	M_2 f = 1.131 I_{leak} = 2.958	P_3 f = 1.125 I_{leak} = 3.015
M_{12} f = 1.097 I_{leak} = 2.219	P_{13} f = 1.102 I_{leak} = 2.308	P_{12} f = 1.100 I_{leak} = 2.268	M_{13} f = 1.103 I_{leak} = 2.341
M_4 f = 1.071 I_{leak} = 1.780	P_5 f = 1.073 I_{leak} = 1.810	P_4 f = 1.074 I_{leak} = 1.834	M_5 f = 1.076 I_{leak} = 1.851
P_{10} f = 1.046 I_{leak} = 1.450	M_{11} f = 1.048 I_{leak} = 1.468	M_{10} f = 1.049 I_{leak} = 1.481	P_{11} f = 1.049 I_{leak} = 1.490
P_6 f = 1.022 I_{leak} = 1.196	M_7 f = 1.024 I_{leak} = 1.207	M_6 f = 1.024 I_{leak} = 1.214	P_7 f = 1.025 I_{leak} = 1.217
M_8 f = 1.000 I_{leak} = 1.000	P_9 f = 1.000 I_{leak} = 1.006	P_8 f = 1.001 I_{leak} = 1.009	M_9 f = 1.001 I_{leak} = 1.009

Values normalized to this tile

Herbert et al., Mitigating the Impact of Variability on Chip-Multiprocessor Power and Performance, TVLSI, 2009

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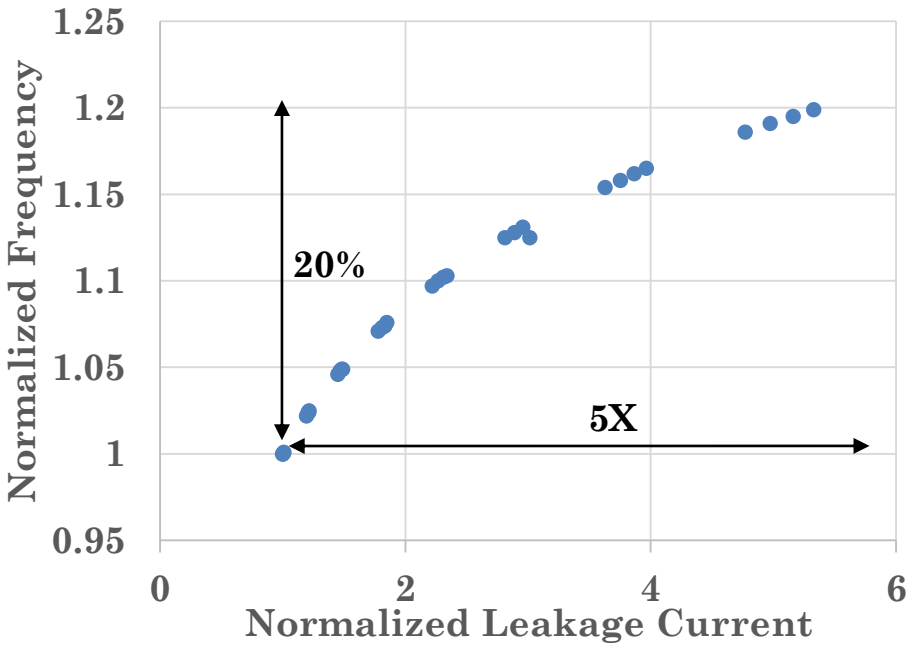
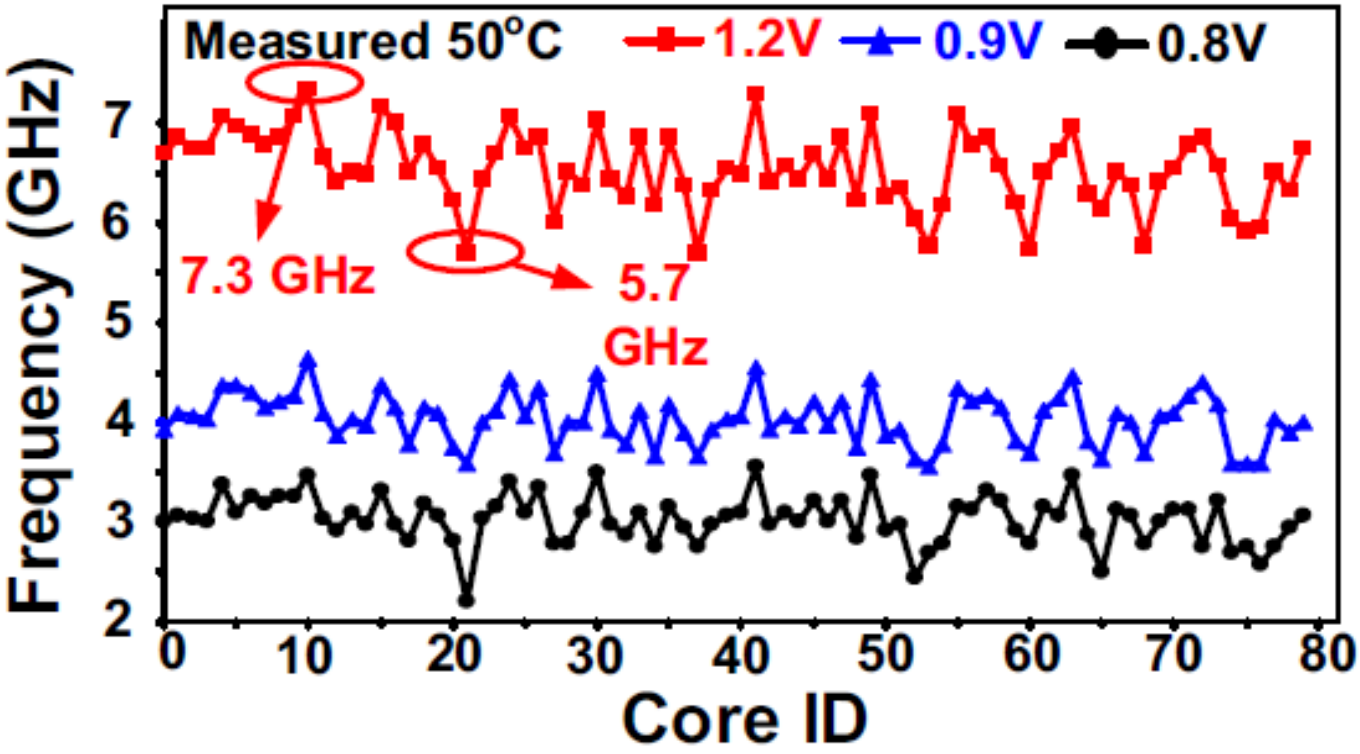


PV can cause large variation in performance and power profile

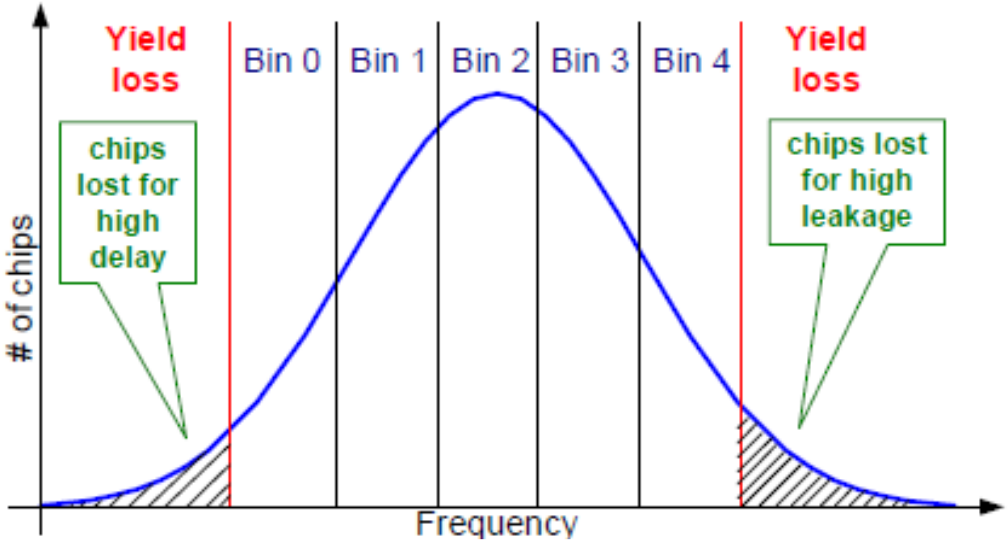
PV in Intel's 80-core processor



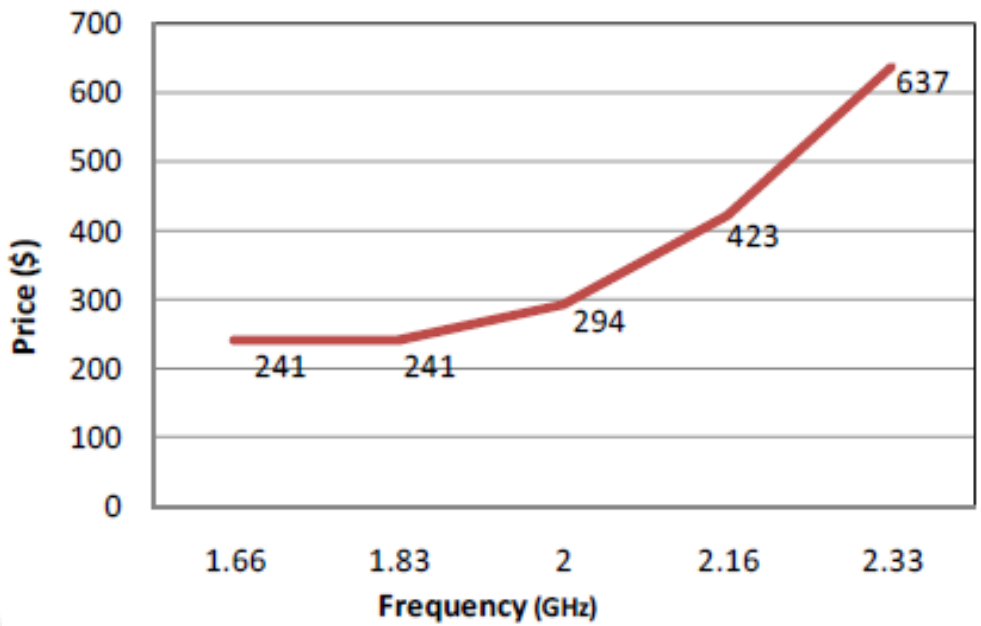
Core frequencies vary widely!

Dighe et al. JSSC'11

Yield & revenue loss due to PV



An example of frequency binning

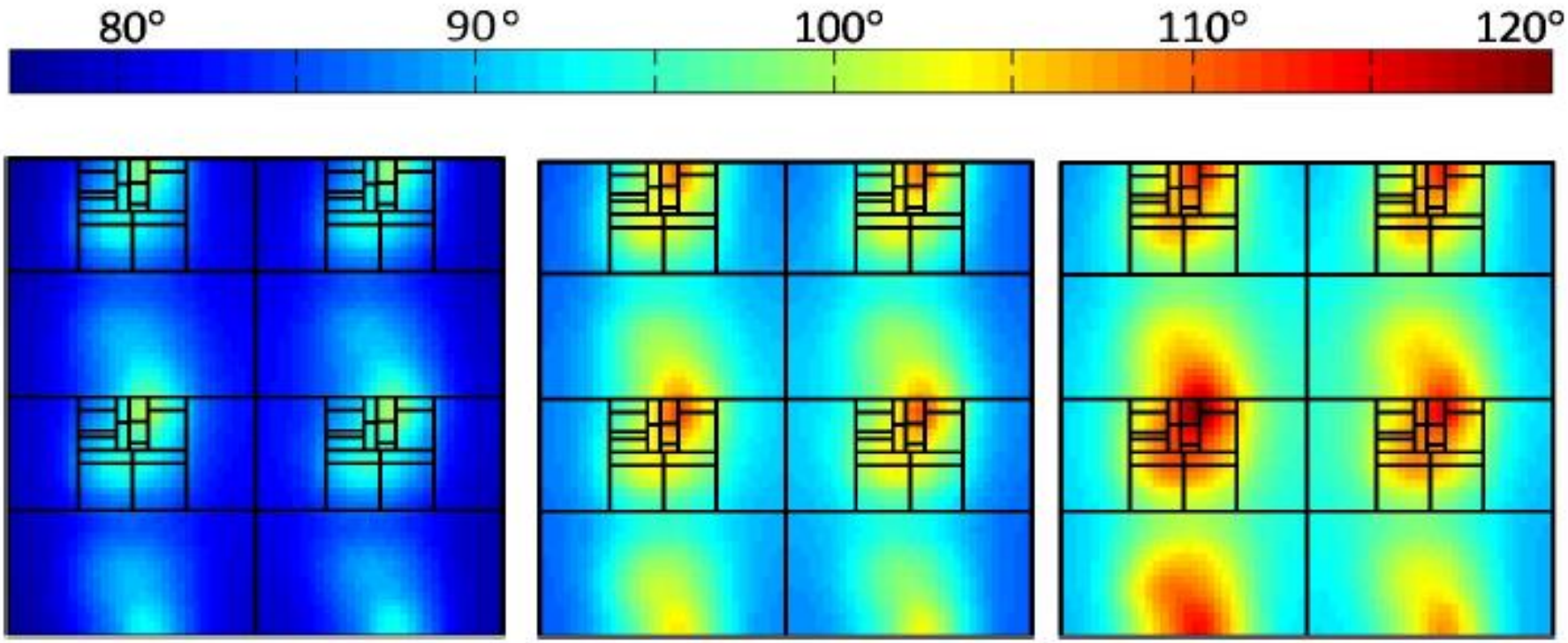


Price vs. frequency of Intel Core Duo mobile processors (Jan'08)

More processors falling in lower bins ➔ loss of revenue

Das et al. MICRO'08

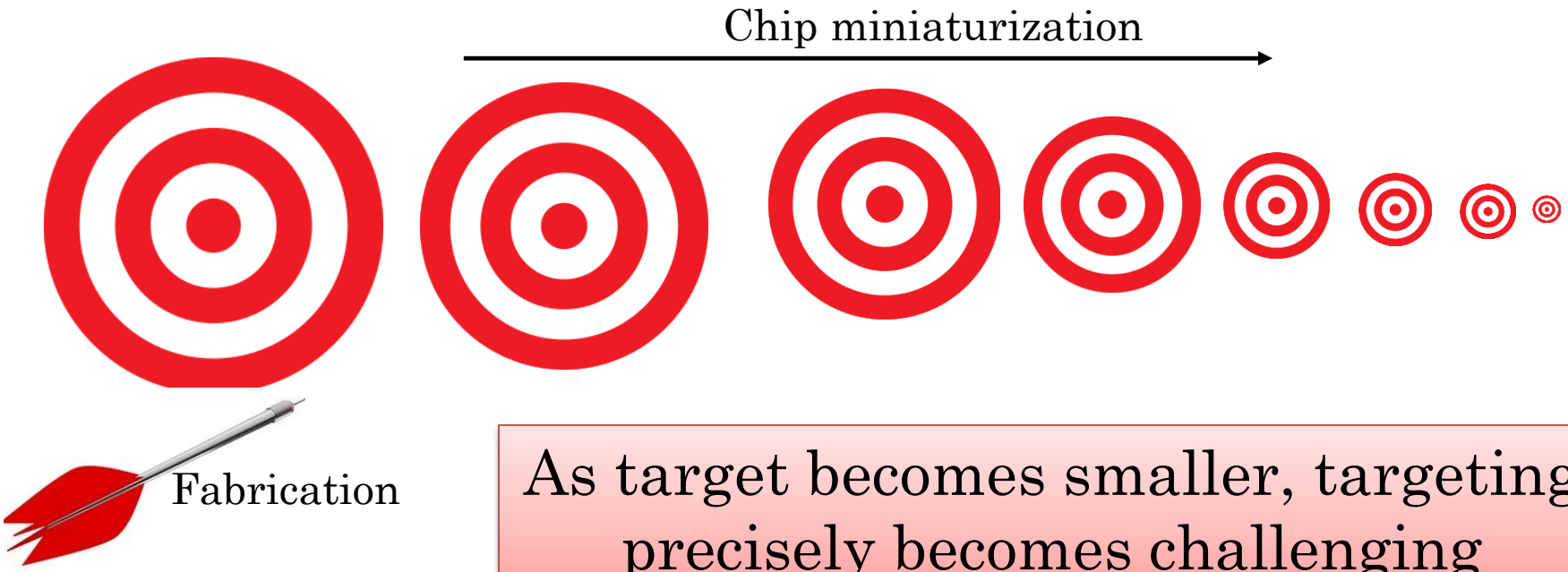
Temperature maps of top tier in a 3D CMP



(a) nominal (b) mild variation (c) severe variation

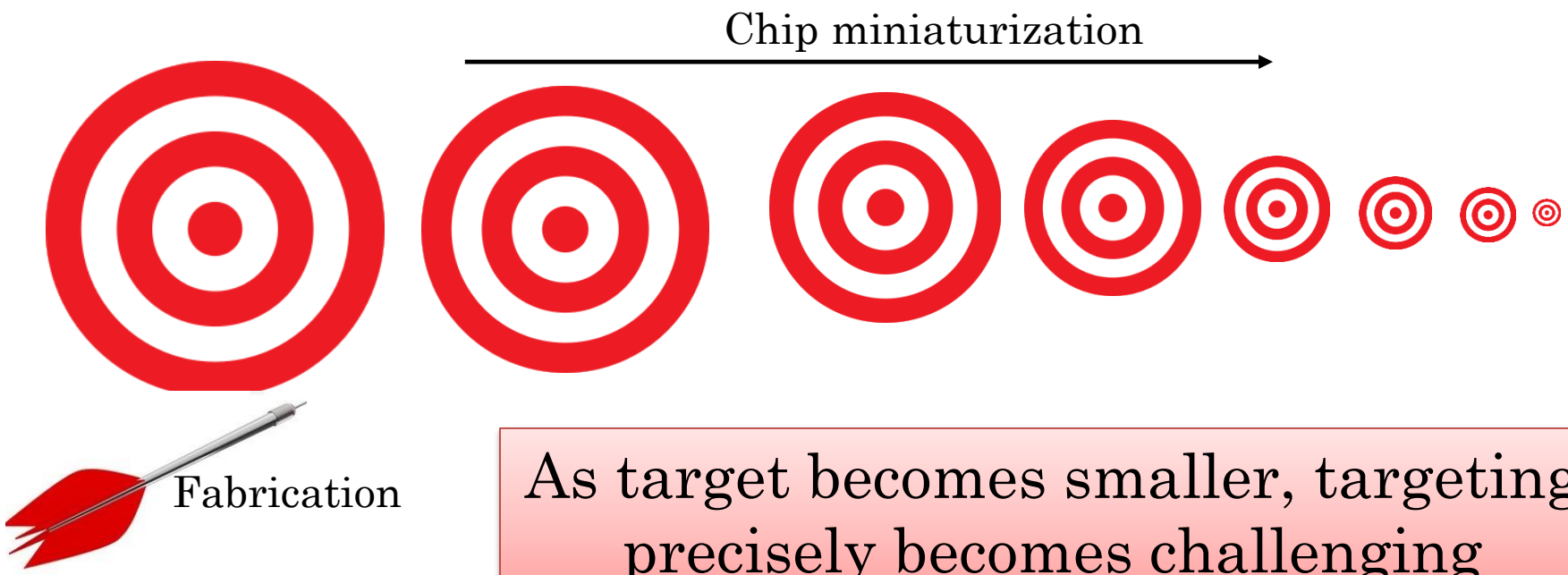
Juan et al. DATE'11

Increasing severity of PV with chip miniaturization



As target becomes smaller, targeting precisely becomes challenging

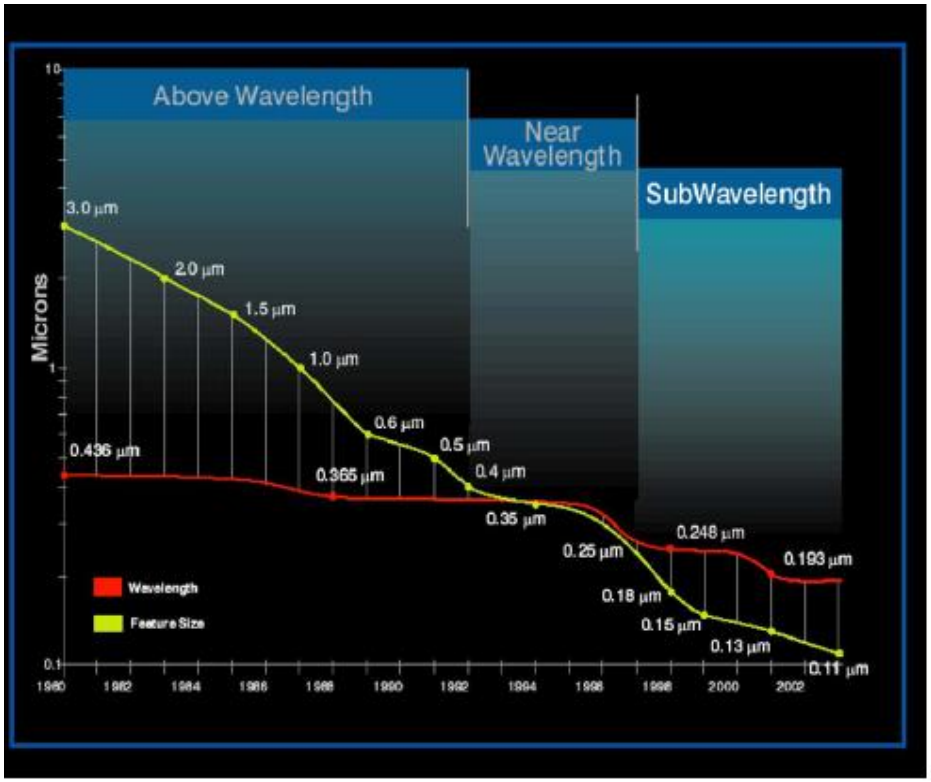
Increasing severity of PV with chip miniaturization



As target becomes smaller, targeting precisely becomes challenging

A crane can accurately displace a ship but not a small pen!

Increasing severity of PV with chip miniaturization



Scaling trend of device feature size and optical wavelength of lithography process (Source: Synopsys Inc.)

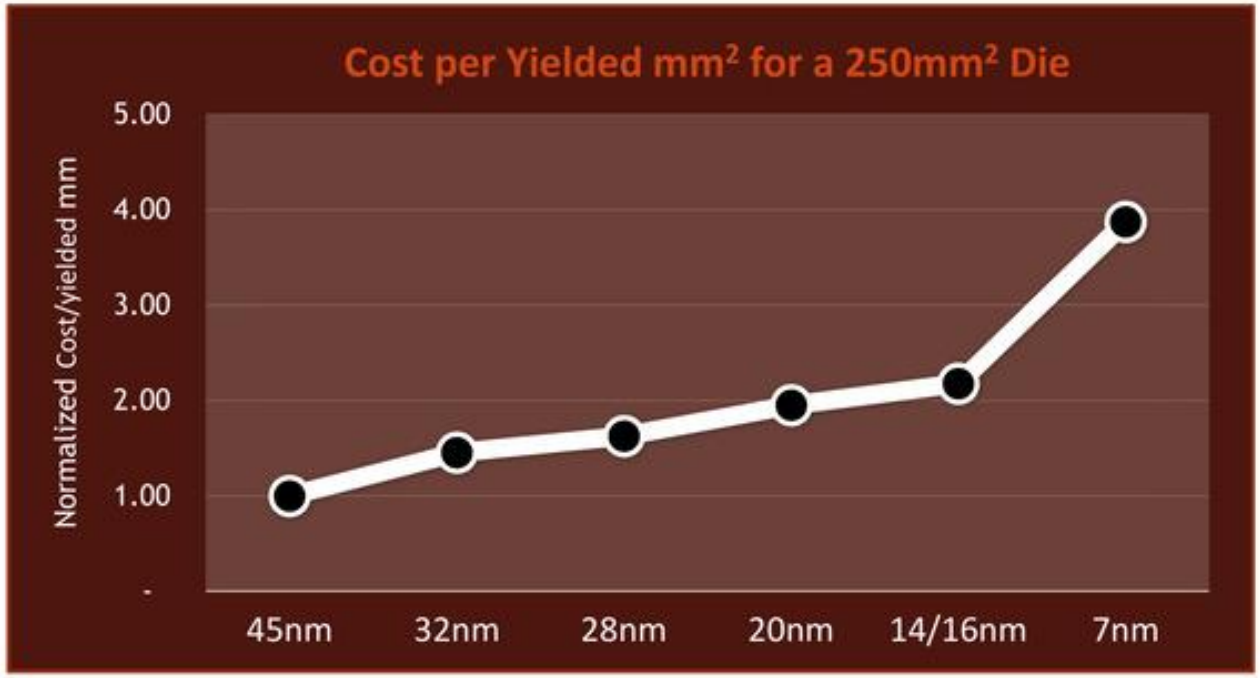
- PV was small above 350nm

However…

- Below 350nm, the feature sizes < wavelength of light

- Printing the layout correctly has become very difficult.

MOORE's LAW Slows While Costs Continue to Increase



INCREASING DIE SIZES ARE ECONOMICALLY PROBLEMATIC

Source: AMD

Granularities and manifestation of PV

- Die-to-die (D2D) variation
- Within-die (WID) variation

- Variation in latency
- Variation in power consumption
- Variation in vulnerability
  - Retention period in eDRAM/DRAM (e.g., 64ms-1sec)
  - Failure probability in SRAM (e.g., 10^{-6} to 10^{-2})
  - Write endurance in non-volatile memories (e.g., 10^{9}-10^{8})

PV affects all ranges of processors, components and parameters

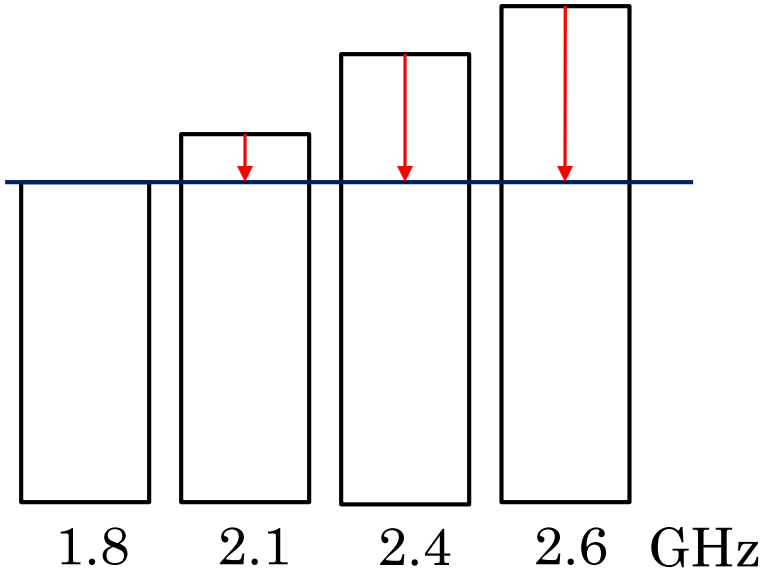
- Maximum clock frequency of different cores in an 80-core Intel processor vary between 5.7 to 7.3 GHz
- 9X variation in sleep power in different instances of ARM Cortex M3 processors.
- 50X variation in write endurance of cells of phase change memory
- Timing parameters in a DDR3 DRAM can be 66% lower than the datasheet specifications
- 15% difference in energy between nodes of the Eurora supercomputer.

Key Ideas of Architectural Management Strategies for Process Variation

Summary of Key Ideas

1. Allocating higher resources to PV-hit parts
2. Reducing burden on PV-hit parts
3. Disabling PV-hit parts
4. Fine-grain (and not coarse-grain) management
5. Extra protection at lower voltages
6. Data mapping based on criticality

Idea 1: Allocating higher resources to PV-hit parts

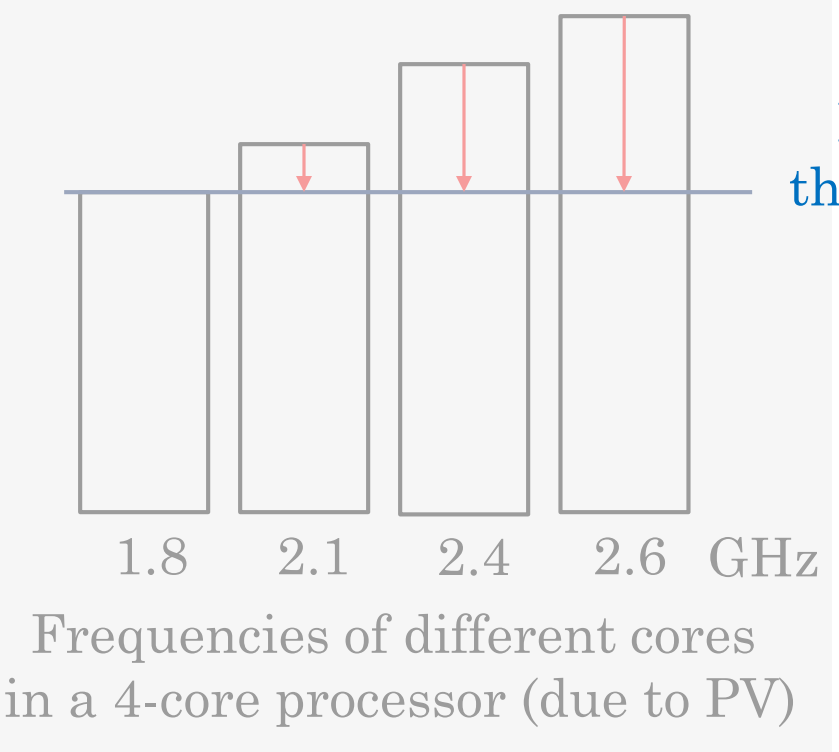


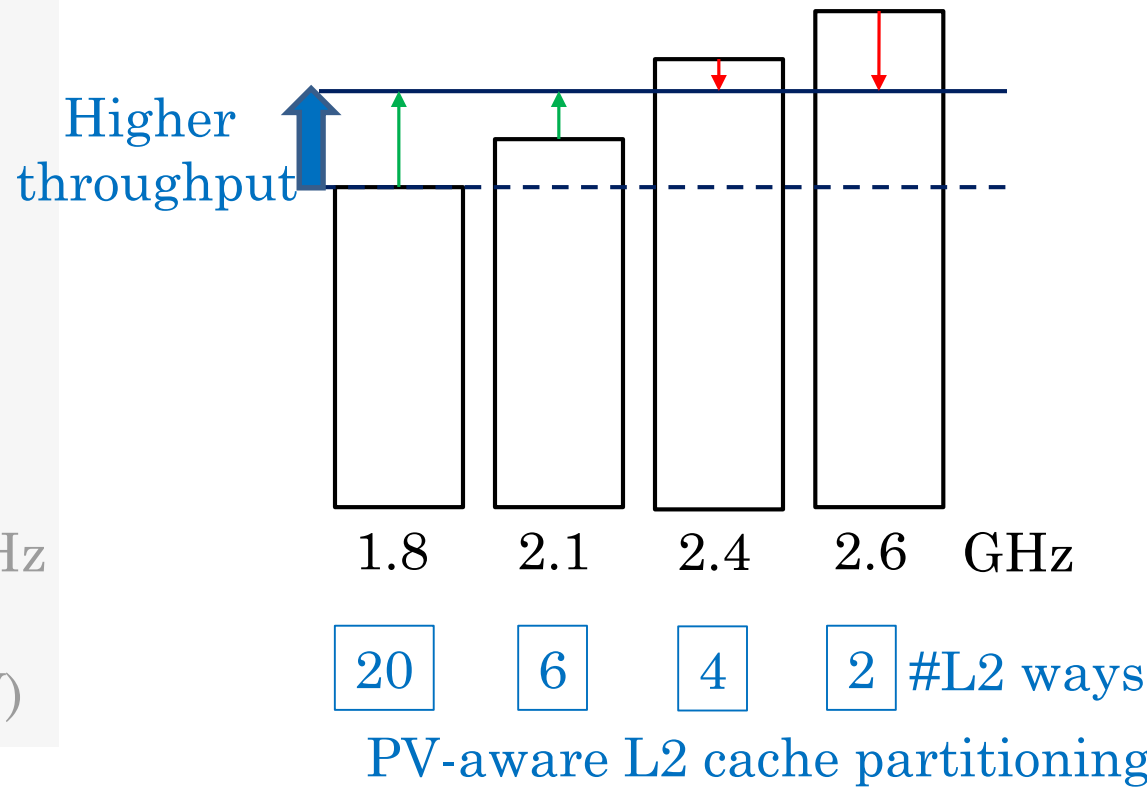
Frequencies of different cores in a 4-core processor (due to PV)

For multithreaded programs with synchronization barriers, the slowest core will limit the performance of other cores

Idea 1: Allocating higher resources to PV-hit parts

- Using cache partitioning to give higher cache quota





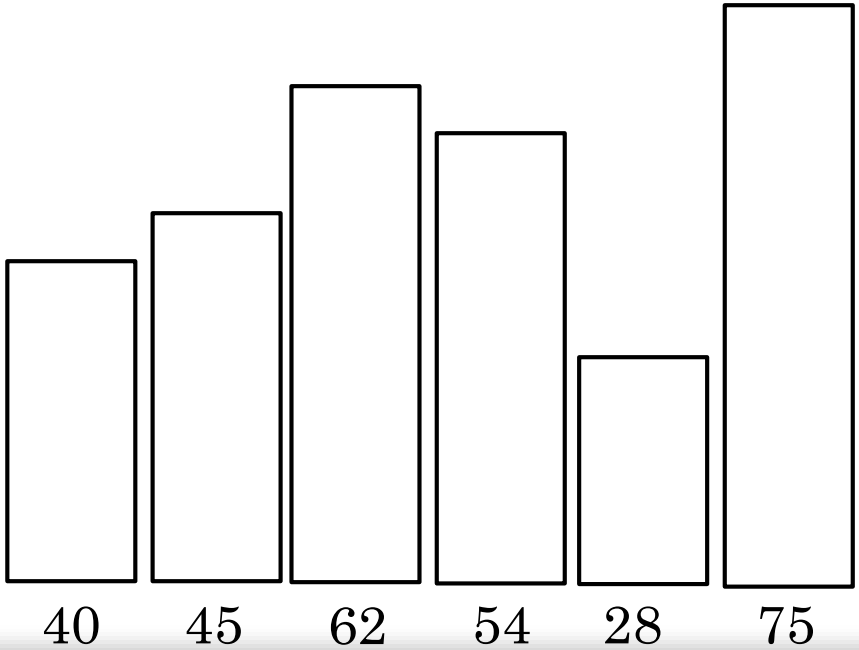
Idea 1: Allocating higher resources to PV-hit parts

- Using cache partitioning to give higher cache quota
- Performing extra refresh operations for (e)DRAM
- Using dynamic voltage/frequency scaling (DVFS)
- Pipeline adaptation to allow extra slack

Idea 2: Reducing burden on PV-hit parts

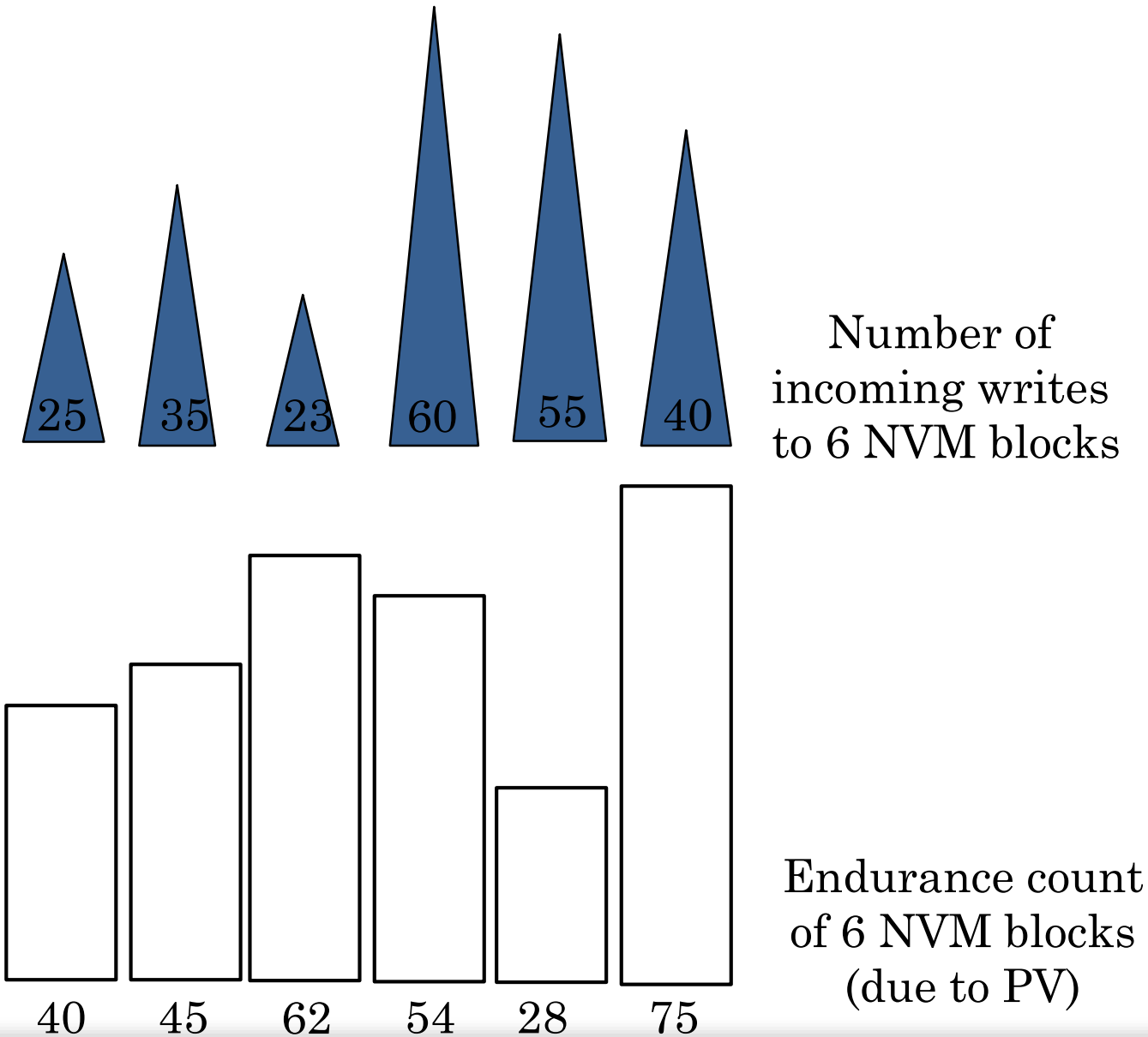
- By directing smaller number of writes to PV-hit NVM blocks, their further degradation can be avoided

Example: write-redistribution in NVM memory



Endurance count of 6 NVM blocks (due to PV)

Example: write-redistribution in NVM memory



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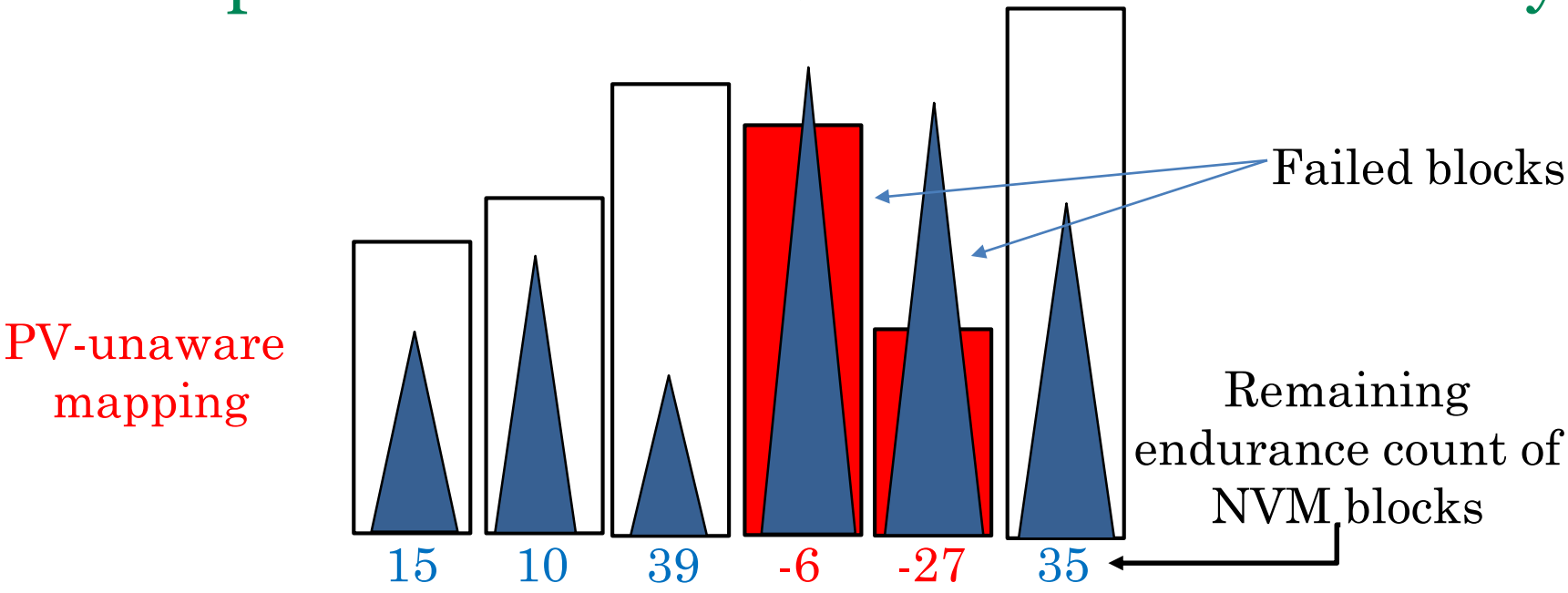

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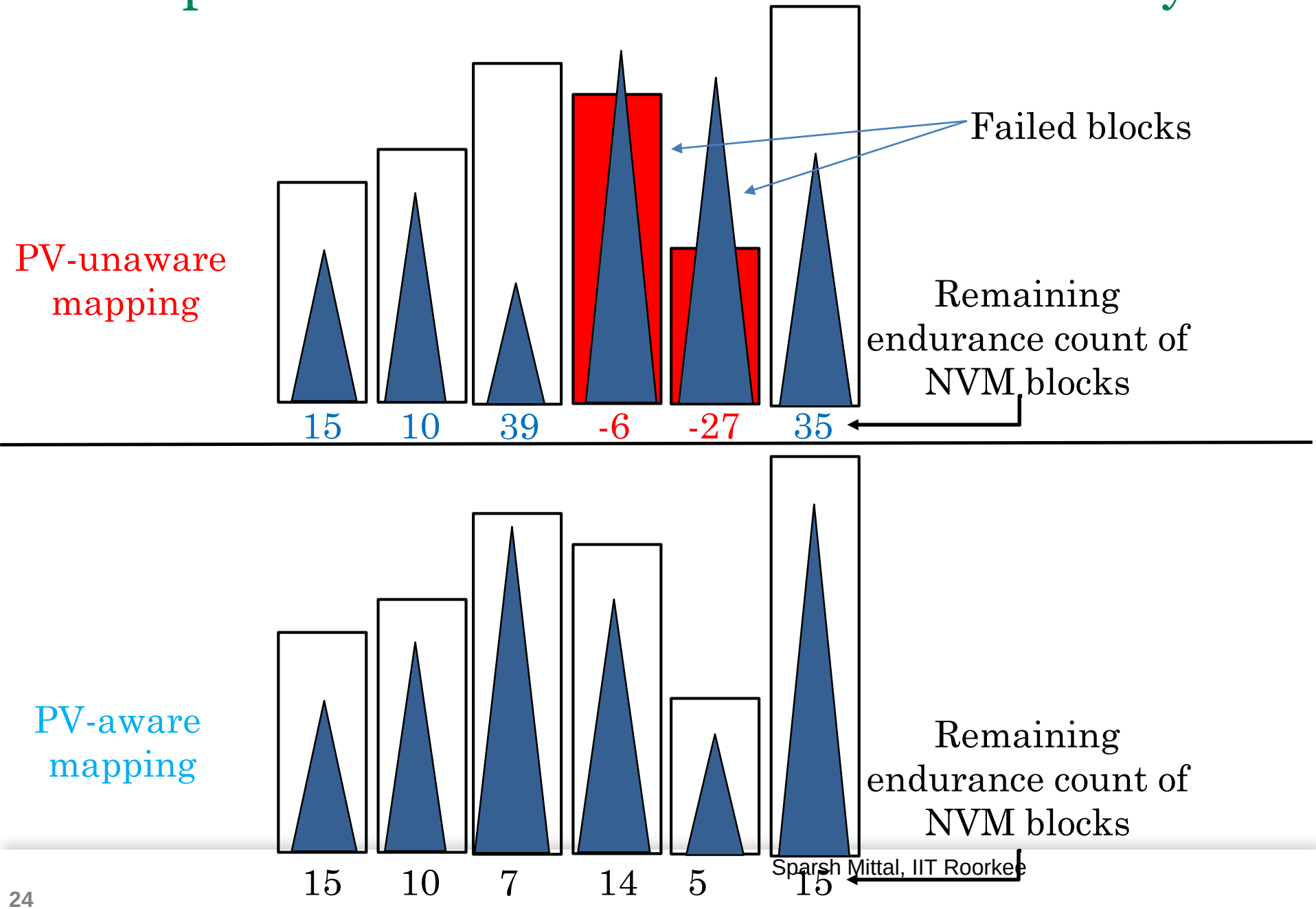


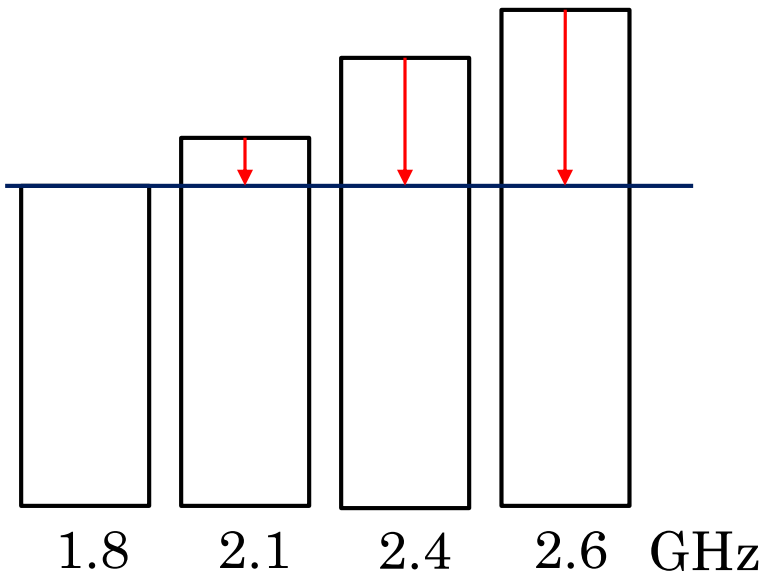
Idea 2: Reducing burden on PV-hit parts

- By directing smaller number of writes to PV-hit NVM blocks, their further degradation can be avoided

- Computationally intensive threads can be mapped to
  - Slower cores for optimizing fairness or
  - Faster cores for optimizing throughput

Idea 3: Disabling PV-hit parts

- PV-hit parts usually limit performance of other parts

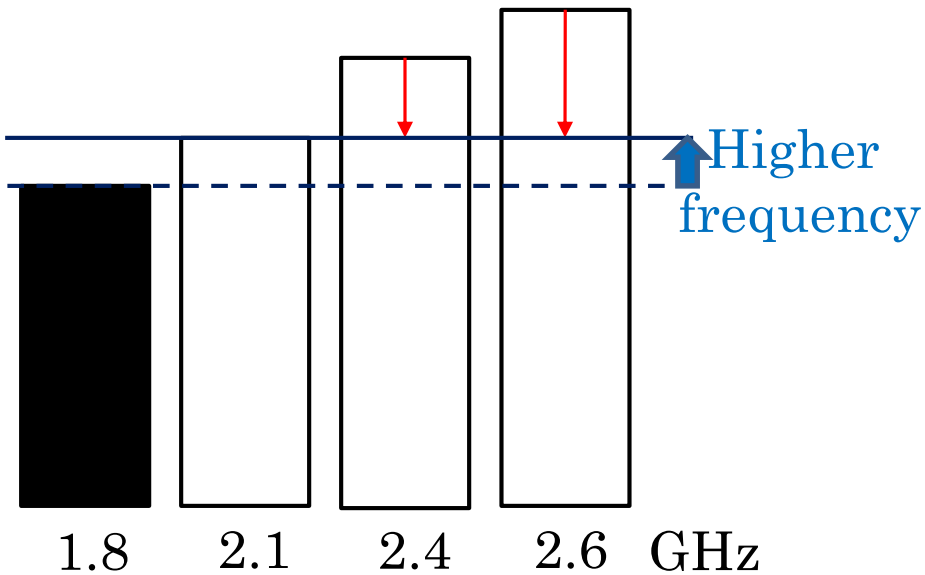


Frequencies of different cores in a 4-core processor (due to PV)

The PV-affected (slowest) core limits the performance of other cores

Idea 3: Disabling PV-hit parts

- Slowest/leakiest/faulty core, cache block, register or functional unit can be disabled



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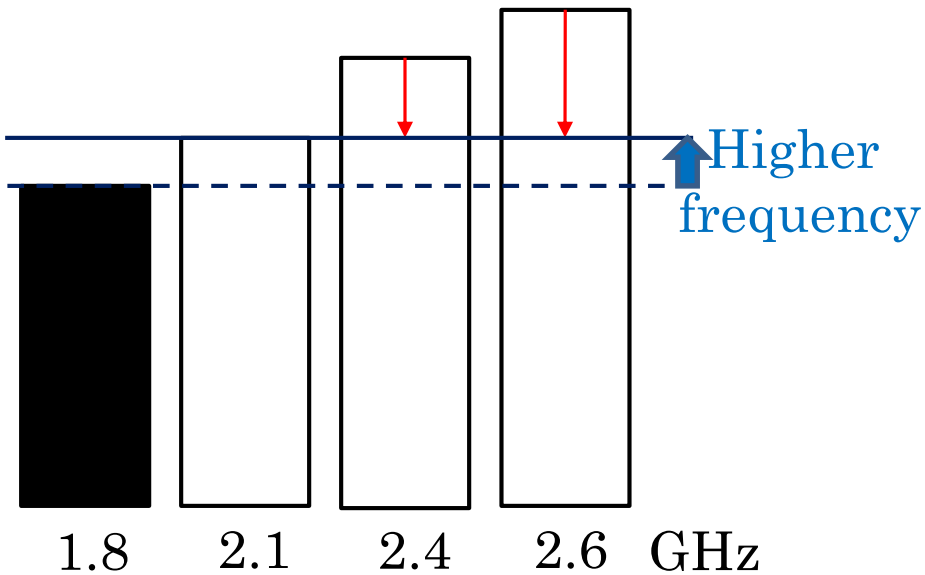
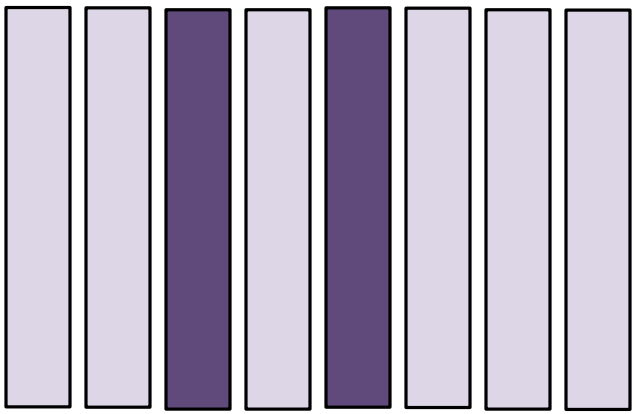


Frequencies of different cores in a 4-core processor (due to PV)

Disabling core may NOT harm performance e.g., for applications with limited parallelism in GPUs.

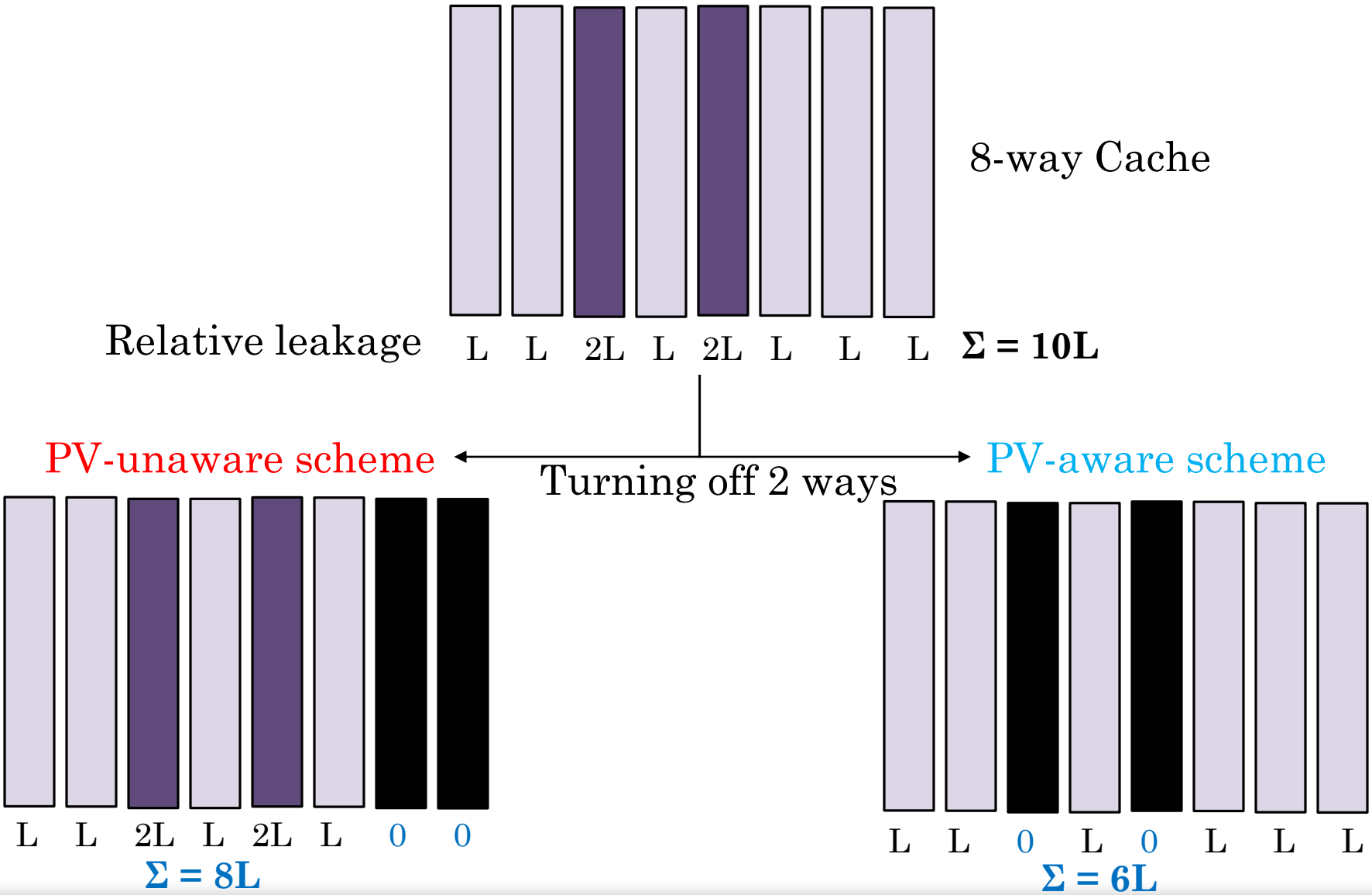
Example: PV-aware cache reconfiguration



8-way Cache

Relative leakage L L 2L L 2L L L L **Σ = 10L**

Example: PV-aware cache reconfiguration



Idea 3: Disabling PV-hit parts

- Disabling high-leakage/latency core and cache ways
- In (e)DRAM, cells with low retention can be disabled
- In NVM, blocks with low endurance can be disabled
- Functionality of faulty computational units can be achieved by memory based computing.
- Errors can be tolerated by approximate computing

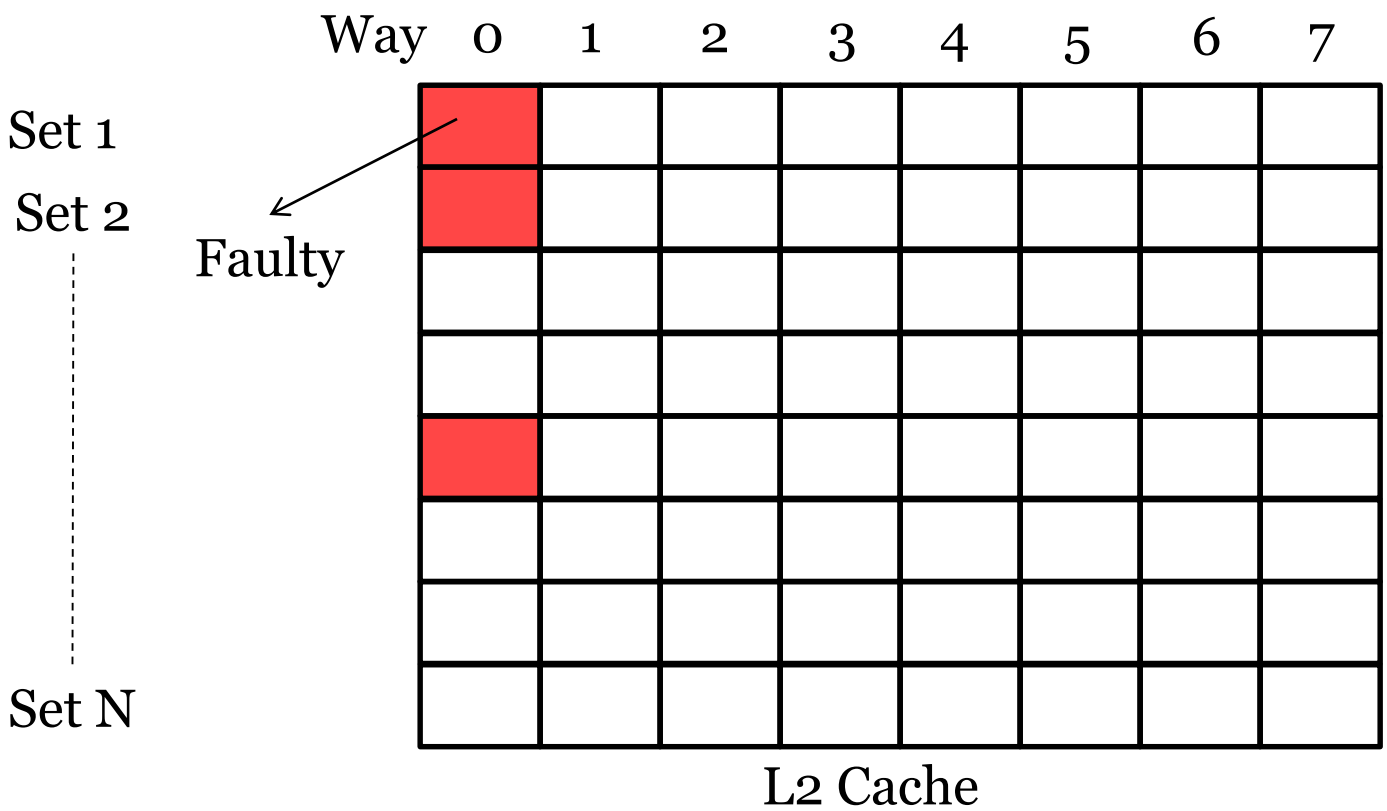
Idea 4: Fine-grain (and not coarse-grain) management

- Observation: not all constituent components are affected by PV, e.g. in a 64B cache block, only few cell may be PV-affected.

- Idea: Instead of deactivating at coarse-grain, we can deactivate at fine-grain and provision just the right amount.

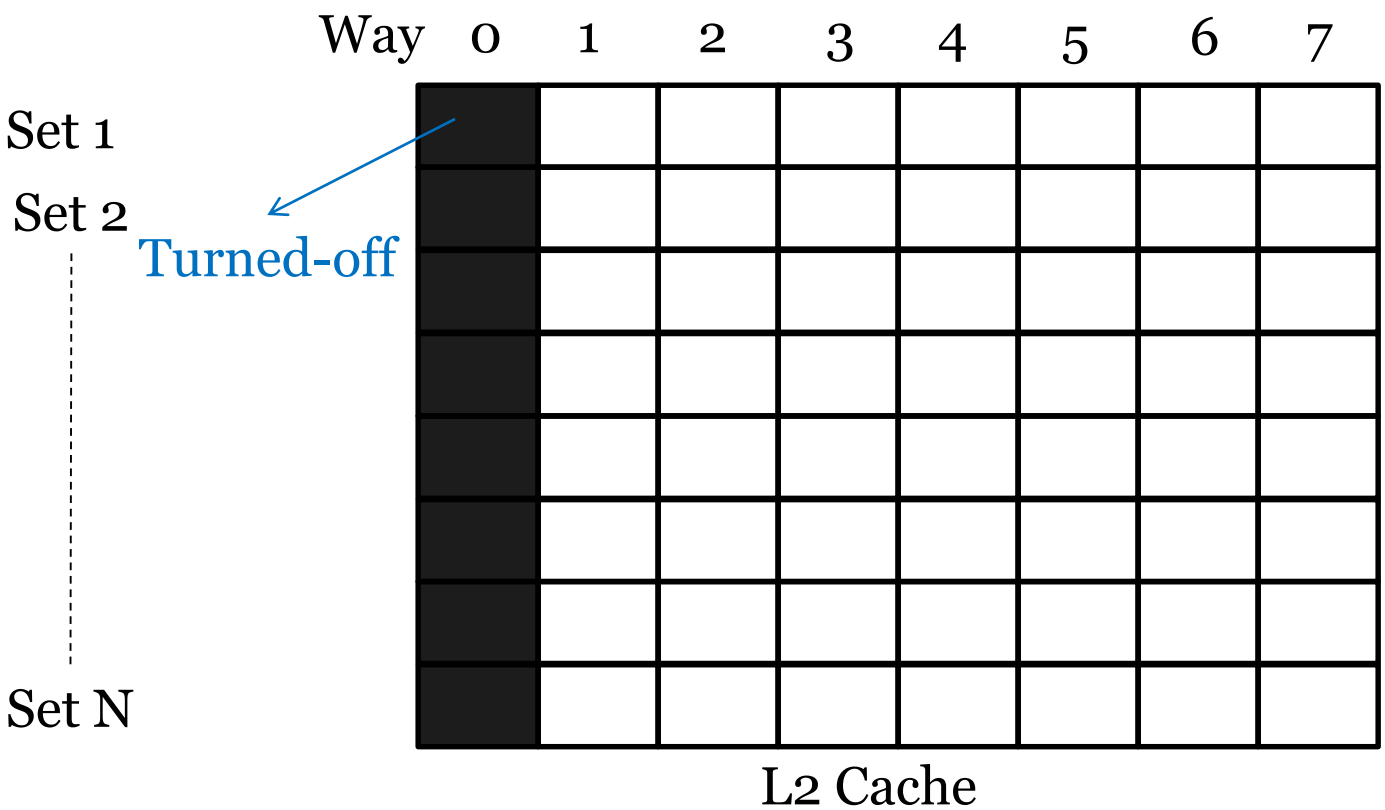
Example: Disabling faulty blocks

- A cache with some faulty blocks (due to PV)



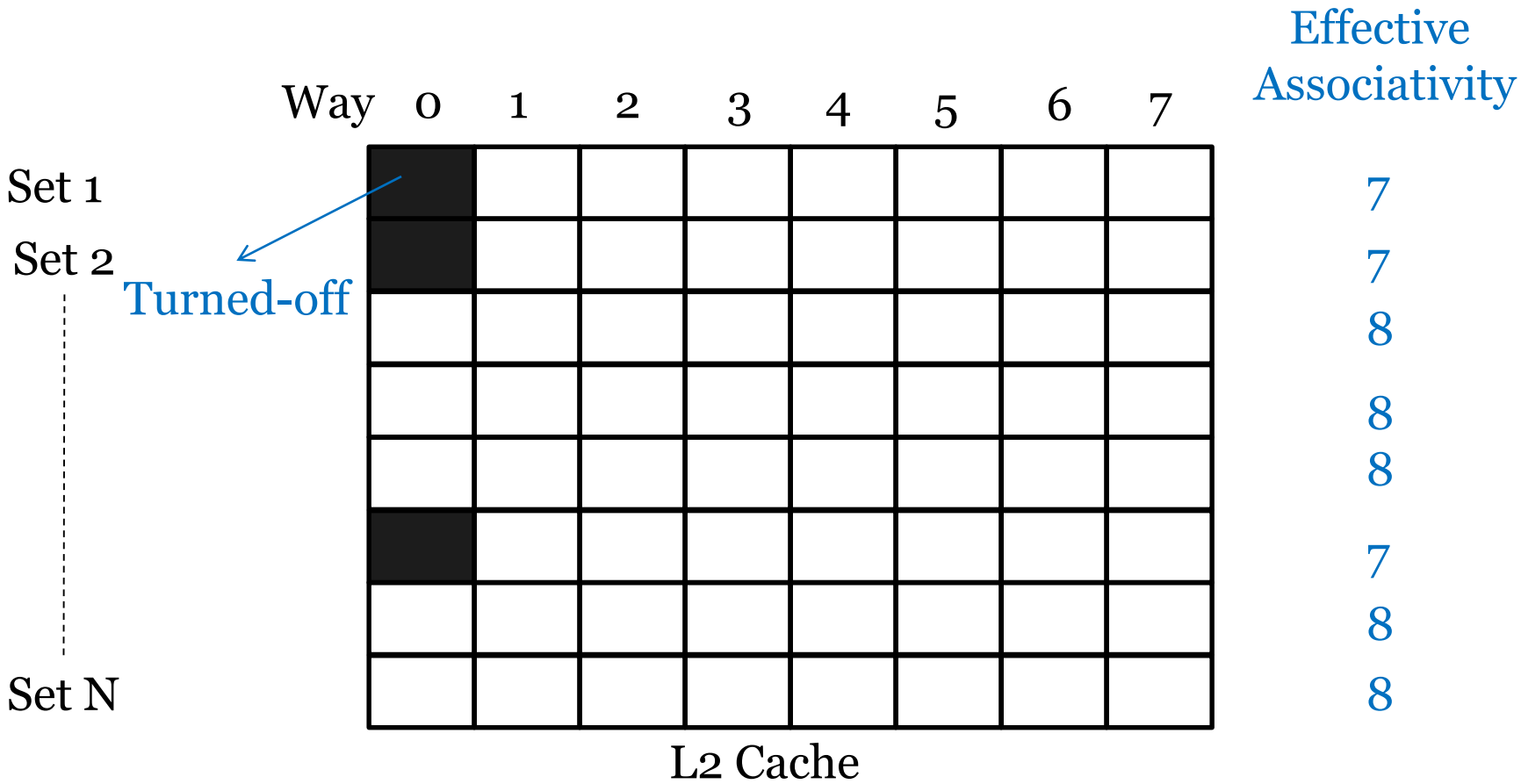
Example: Disabling faulty blocks

- Naïve coarse-grain approach: disabling entire way



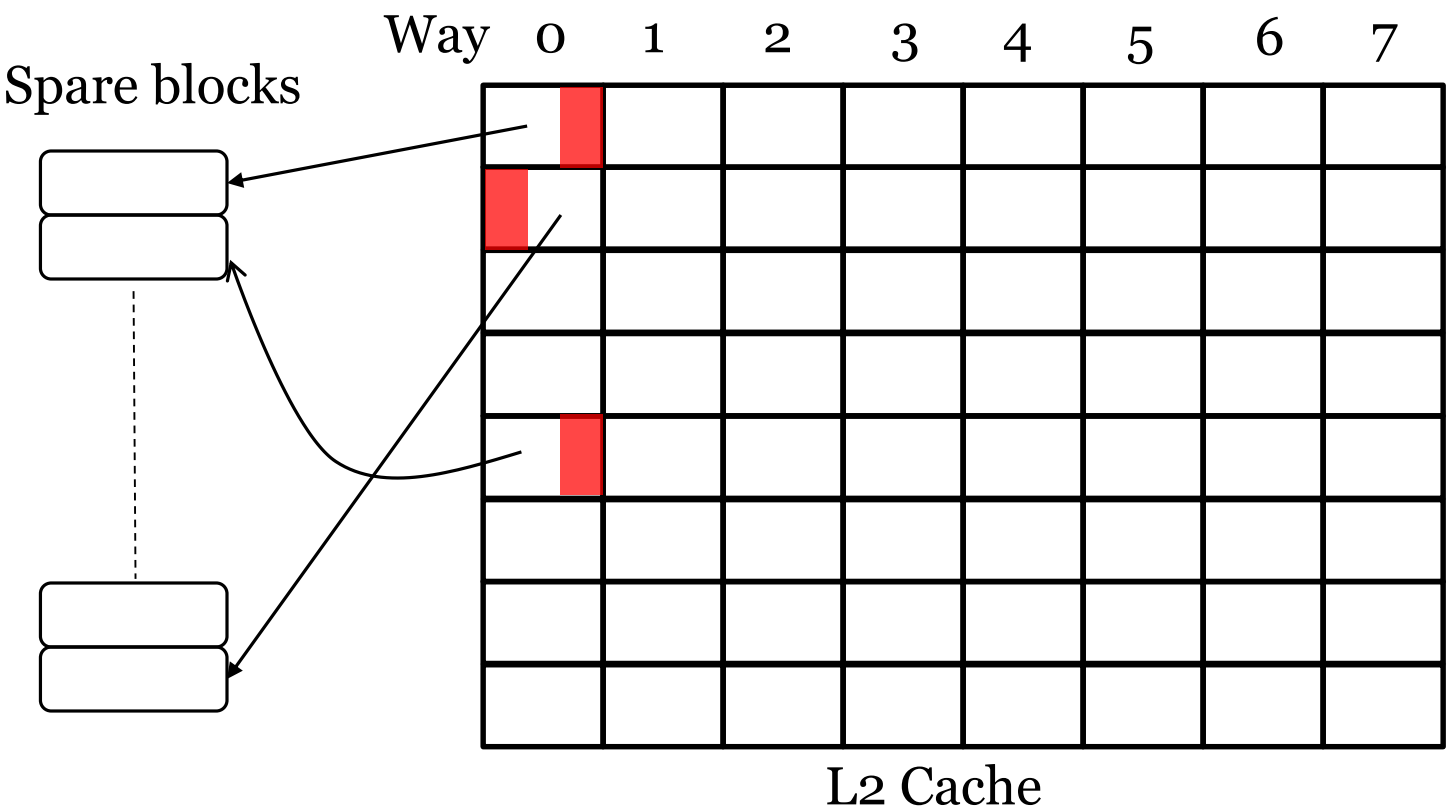
Example: Disabling faulty blocks

- Intelligent approach 1: disabling only faulty blocks



Example: Disabling faulty blocks

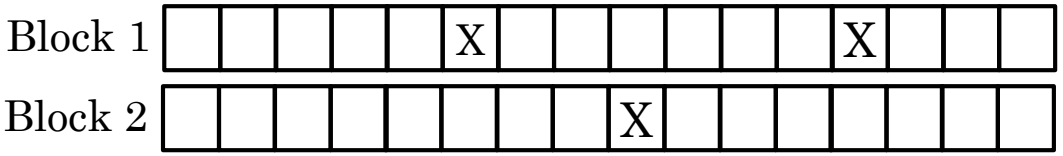
- Intelligent approach 2: redirecting accesses to spare



Idea 4: Fine-grain (and not coarse-grain) management

- Providing extra refresh or stronger ECC only for weak cells and not all the cells

- Using multiple faulty blocks as a single block, if faulty subblocks are at different positions



Idea 5: Extra protection at lower voltages

- Effect of PV is increased at lower voltages

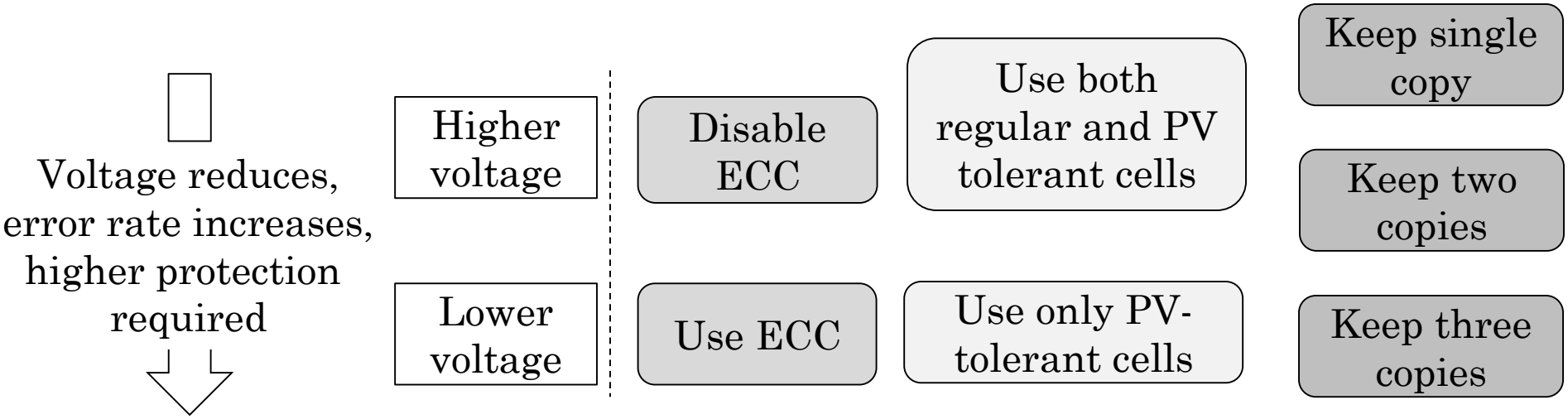
Voltage reduces, error rate increases, higher protection required

Higher voltage

Lower voltage

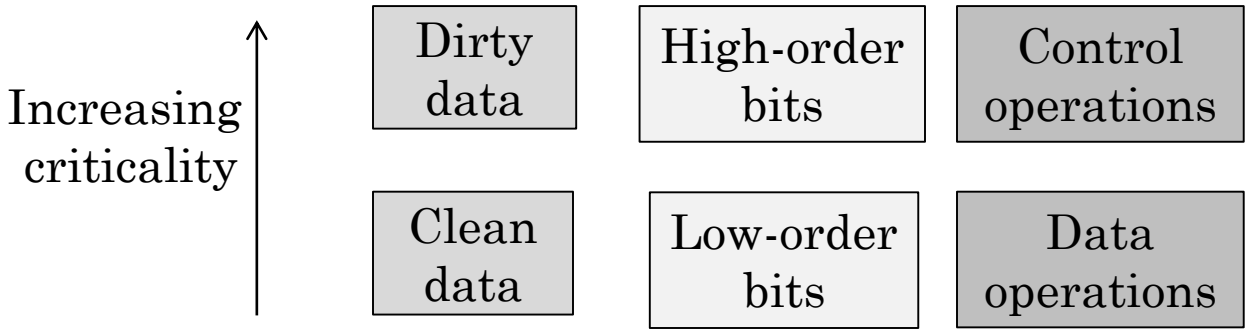
Idea 5: Extra protection at lower voltages

- Higher reliability resources can be provided at lower voltages



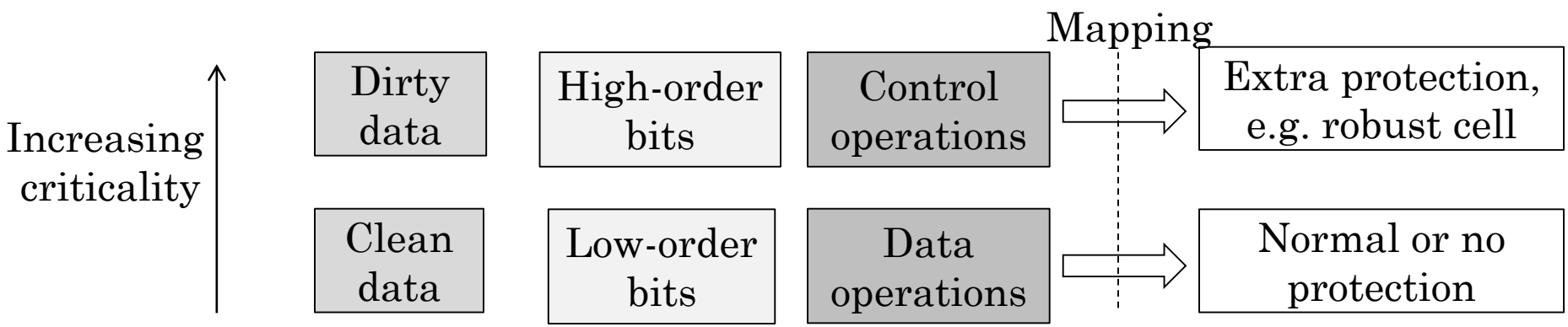
Idea 6: Data mapping based on criticality

- Some regions/instructions have higher criticality



Idea 6: Data mapping based on criticality

- Higher protection can be provided to more critical or more vulnerable portions.



References

- S. Mittal, "A Survey Of Architectural Techniques for Managing Process Variation", Computing Surveys, 2016 (pdf)
- S. Mittal et al., “A Survey of Techniques for Modeling and Improving Reliability of Computing Systems”, IEEE TPDS, 2016 (pdf)
- S. Ghosh et al. "Parameter variation tolerance and error resiliency: New design paradigm for the nanoscale era", Proc. IEEE, 2010
- Borkar et al. “Parameter variations and impact on circuits and microarchitecture.”, DAC, 2003
- Sparsh Mittal, "A Survey Of Architectural Techniques for Near-Threshold Computing", ACM JETC 2015. (pdf)